

Local-global methods for elliptic curves

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- 1 Elliptic curves
- 2 The Birch and Swinnerton-Dyer conjecture
- 3 Using 'local' information to control 'global' data

Global: $\mathbb{Q}$
Hard

Local: $\mathbb{R}, \mathbb{F}_p$
Accessible

- 4 Results

Solving polynomial equations

Goal: Describe the rational solutions to $f(x, y) = 0$.

$y - 2x + 1 = 0$	$x = t, y = 2t - 1$ for $t \in \mathbb{Q}$	very easy
$x^2 + y^2 - 1 = 0$	$x = \frac{2t}{1+t^2}, y = \frac{1-t^2}{1+t^2}$ for $t \in \mathbb{Q}$	easy
$x^2 + y^2 - 3 = 0$	no rational solutions	easy
$x^3 - y^2 - x + 1 = 0$	$x = 0, y = \pm 1, \quad x = 1, y = \pm 1, \quad ???$	hard!

Quadratics satisfy the Hasse principle:

Theorem (Hasse–Minkowski, 1920s)

Let f be a homogeneous quadratic equation. Then,

f has a $\mathbb{Q}$ solution $\iff$ f has an $\mathbb{R}$ solution and $f \bmod p^k$ has a solution
for all primes p and all $k \geq 1$.

Cubics don't satisfy the Hasse principle ☹, but not all is lost!

Elliptic curves

Fix a cubic $f(x) \in \mathbb{Q}[x]$ with $\text{Disc}(f) \neq 0$. Consider

$$E : y^2 = f(x).$$

An **elliptic curve** is the graph of such an equation.

Solutions to the equation $\rightsquigarrow$ points on the curve.

Theorem (Mordell, 1922)

$E(\mathbb{Q})$ is a finitely generated abelian group, i.e.,

$$E(\mathbb{Q}) \cong E(\mathbb{Q})_{\text{tors}} \oplus \mathbb{Z}^{r_E}$$

for some finite group $E(\mathbb{Q})_{\text{tors}}$ and $r_E \in \mathbb{N}$.

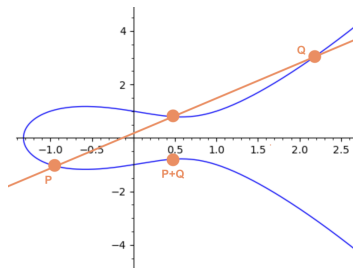
$E(\mathbb{Q})_{\text{tors}}$ is well studied:

Theorem (Mazur, 1977)

There are 15 possibilities for $E(\mathbb{Q})_{\text{tors}}$.

Much less is known about r_E ...

- Is r_E bounded/unbounded?
- An algorithm to compute r_E for all curves?



$$E : y^2 = x^3 - x + 1$$

Have $E(\mathbb{Q})_{\text{tors}} = 1$, $r_E = 1$.

In particular, $E(\mathbb{Q}) = \mathbb{Z} \cdot (1, 1)$.

Relating $E(\mathbb{Q})$ to $E(\mathbb{F}_p)$

Key question: What information does $E(\mathbb{F}_p)$ give us about $E(\mathbb{Q})$?

- A naïve counting argument gives:

$$\#E(\mathbb{F}_p) \approx p + 1.$$

- Hasse (1933) proved:

$$\underbrace{|\#E(\mathbb{F}_p) - (p + 1)|}_{a_p} \leq 2\sqrt{p}.$$

- Birch and Swinnerton-Dyer (1960s) observed:

There's a constant C_E such that:

$$\prod_{p \leq X} \frac{\#E(\mathbb{F}_p)}{p} \sim C_E \log(X)^{r_E} \text{ as } X \rightarrow \infty.$$

Definition (L -function attached to E)

$$L(E, s) \text{ "=" } \prod_p \frac{1}{1 + a_p p^{-s} + p^{1-2s}}$$

is a complex analytic function defined when $\operatorname{Re}(s) > \frac{3}{2}$.

Note that,

$$L(E, 1) \text{ "=" } \prod_p \frac{p}{\#E(\mathbb{F}_p)}.$$

The Birch and Swinnerton-Dyer conjecture

Let E be an elliptic curve. Then $E(\mathbb{Q}) \cong E(\mathbb{Q})_{\text{tors}} \oplus \mathbb{Z}^{r_E}$.

Conjecture (Birch–Swinnerton-Dyer I)

0 $L(E, s)$ has an analytic continuation to $\mathbb{C}$,

1 $r_E = \text{ord}_{s=1} L(E, s)$.

If BSD is true, then there is an algorithm to compute r_E and $E(\mathbb{Q})$!

Current status

0 follows from the modularity theorem (2001),

1 known to hold:

- when $\text{ord}_{s=1} L(E, s) \leq 1$ (2001),
- *modulo 2 (2010),
- for $> 66\%$ of all elliptic curves (2014).

Conjecture (Birch–Swinnerton-Dyer II)

$L(E, s) = C_E(s-1)^{r_E} + \text{higher order terms}$ where,

$$C_E = \frac{\text{Reg}_E \cdot \#\text{III}_E \cdot \Omega_E \cdot \prod_p c_{p,E}}{\#E(\mathbb{Q})_{\text{tors}}^2}.$$

Current status: ■ Some progress when $\text{ord}_{s=1} L(E, s) \leq 1$.

■ When $\text{ord}_{s=1} L(E, s) > 1$, don't even know that $\#\text{III}_E < \infty \dots$

The Birch and Swinnerton-Dyer invariants

Conjecture (Birch–Swinnerton-Dyer II)

Let E be an elliptic curve. Then $L(E, s) = C_E(s-1)^{r_E} + \text{higher order terms}$ where,

$$C_E = \frac{\text{Reg}_E \cdot \#\text{III}_E \cdot \Omega_E \cdot \prod_p c_{p,E}}{\#E(\mathbb{Q})_{\text{tors}}^2}.$$

Regulator

Write $E(\mathbb{Q})/E(\mathbb{Q})_{\text{tors}} \cong \mathbb{Z} \cdot P_1 \oplus \cdots \oplus \mathbb{Z} \cdot P_{r_E}$,
then

$$\text{Reg}_E = \det(\langle P_i, P_j \rangle) \in \mathbb{R}$$

is the squared covolume of the lattice.

Tate–Shafarevich group

III_E measures the how badly E fails the
Hasse principle.

It's only conjectured that III_E is finite!
If III_E is finite, it has $\square$ order (Cassels).

Real period

$$E(\mathbb{C}) \cong \text{torus} \cong \mathbb{C}/\Lambda.$$

$$\Omega_E = \text{covol}(\Lambda \cap \mathbb{R}) \times \#\text{Comp}(E(\mathbb{R})) \in \mathbb{R}$$

measures the 'length' of E over $\mathbb{R}$.

Tamagawa number at p

$c_{p,E} \in \mathbb{N}$ records how 'bad' E looks modulo p .

- $y^2 = (x-r_1)(x-r_2)(x-r_3) \rightsquigarrow$ good, $c_{p,E} = 1$
- $y^2 = (x-r_1)^2(x-r_2) \rightsquigarrow$ bad
- $y^2 = (x-r_1)^3 \rightsquigarrow$ very bad.

Computing the parity of the rank

An isogeny between elliptic curves is a map preserving their algebraic structure.

Let $\phi : E_1 \rightarrow E_2$ be an isogeny and let $d = \# \ker(\phi)$.

Theorem (Cassels, 1965)

Assume that $\#\text{III}_{E_1}$ is finite. Then,
 $C_{E_1} = C_{E_2}$.

Lemma

$$\frac{\text{Reg}_{E_1}}{\text{Reg}_{E_2}} = \square \cdot d^{r_{E_1}}$$

$$\implies \square \cdot d^{r_{E_1}} = \frac{\text{Reg}_{E_1}}{\text{Reg}_{E_2}} = \square \cdot \underbrace{\frac{\Omega_{E_2}}{\Omega_{E_1}}}_{\mathbb{R}} \cdot \prod_p \underbrace{\frac{c_{p,E_2}}{c_{p,E_1}}}_{\mathbb{F}_p}$$

Theorem (Birch–Cassels)

Assume that $\#\text{III}_{E_1}$ is finite and that d is prime. Then,

$$r_{E_1} \equiv \text{ord}_d\left(\frac{\Omega_{E_2}}{\Omega_{E_1}}\right) + \sum_{p \text{ prime}} \text{ord}_d\left(\frac{c_{p,E_2}}{c_{p,E_1}}\right) \pmod{2}.$$

This is a ‘local’ recipe for the parity of r_{E_1} !

$$\begin{aligned} r_{E_1} \text{ odd} &\implies r_{E_1} \neq 0 \\ &\implies E_1(\mathbb{Q}) \text{ is infinite!} \end{aligned}$$

Computing the parity of the rank

There's an isogeny $\phi : E_1 \rightarrow E_2$ with $\# \ker(\phi) = 2$ when:

$$E_1 : y^2 = \overbrace{x^3 - x^2 - 13x + 22}^{f_1(x)},$$
$$E_2 : y^2 = \overbrace{x^3 - x^2 + 12x + 72}^{f_2(x)}.$$

Theorem (Birch–Cassels)

Assuming $\# \text{III}_{E_1}$ is finite,

$$r_{E_1} \equiv \text{ord}_2\left(\frac{\Omega_{E_2}}{\Omega_{E_1}}\right) + \sum_{p \text{ prime}} \text{ord}_2\left(\frac{c_{p,E_2}}{c_{p,E_1}}\right) \pmod{2}.$$

$$r_{E_1} \equiv \text{ord}_2\left(\frac{1.9034\dots}{3.8069\dots}\right) + \sum_{p \neq 2,3,5} \text{ord}_2\left(\frac{1}{1}\right) + \text{ord}_2\left(\frac{5}{5}\right) + \text{ord}_2\left(\frac{2}{2}\right) + \text{ord}_2\left(\frac{2}{2}\right) = -1$$

p	$f_1(x) \bmod p$	$f_2(x) \bmod p$
$p \neq 2, 3, 5$	distinct roots	distinct roots
$p = 3$	$(x-1)^2(x-2)$	$x^2(x-1)$
$p = 5$	$(x-2)^3$	$(x-2)^3$

$$r_{E_1} \text{ odd} \implies r_{E_1} \neq 0 \\ \implies E_1(\mathbb{Q}) \text{ is infinite!}$$

Generalising all of this

Tate later generalised the statement of the Birch and Swinnerton–Dyer conjecture:

Elliptic curves $\rightsquigarrow$ Abelian varieties

$\mathbb{Q}$ $\rightsquigarrow$ Number fields

In this setting, much less is known.

- No longer know that the L -function is defined at $s = 1$.
- Have significantly less numerical evidence.
- $\#III$ not known to be finite, and we have examples where it is not a square.

Results (assuming $\#III$ is finite).

Joint with E. Aylward, V. Dokchitser, A. Konstantinou, A. Morgan

Theorem

The parity of the rank of the Jacobian of a curve can be described explicitly in terms of local data.

Theorem

BSD I holds modulo 2 for:

- elliptic curves,
- Jacobians of hyperelliptic curves*.

Thank you for your attention!